Institute of Management Sciences (IMSciences)

Peshawar, Pakistan

Project Report

IoT-Driven Machine Learning Framework for Fraud Detection and Customer Risk Prediction in Retail Banking

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Prepared By: Usman Ahmad

Assigned By: Dr. Adnan Amin

Department: Computer Science

University: IM Sciences University, Peshawar

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Academic Research Project Submission

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1 Dataset Selection and Description

1.1 Dataset Name

Credit Card Fraud Detection Dataset

1.2 Dataset Source

`https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud`

1.3 Dataset Overview

This project uses the Credit Card Fraud Detection Dataset containing real-world European credit card transactions. The dataset is widely used for fraud detection research because it represents real banking fraud scenarios with highly imbalanced classes.

The target variable is **Class**, where:

- 0 = Legitimate Transaction
- 1 = Fraudulent Transaction

1.4 Dataset Statistics

Table 1: Dataset Summary

Total Samples	284,807
Total Features	30 Input Features + 1 Target Variable
Fraud Cases	492
Non-Fraud Cases	284,315
Fraud Percentage	0.1727%

This severe class imbalance reflects real-world fraud detection challenges in retail banking.

1.5 Preprocessing Steps

1. Feature and target separation
2. Standardization of the `Amount` feature using StandardScaler

3. Train-test split using 80/20 ratio
4. Stratified sampling with random_state = 42
5. Class balancing using SMOTE on training data only

Table 2: Train-Test Split

Training Data	(227,845, 30)
Testing Data	(56,962, 30)

2 Proposed Model Visualization

2.1 System Workflow

Input Data
↓
Data Preprocessing
↓
Feature Engineering
↓
Model Architecture
↓
Training
↓
Evaluation
↓
Output / Prediction

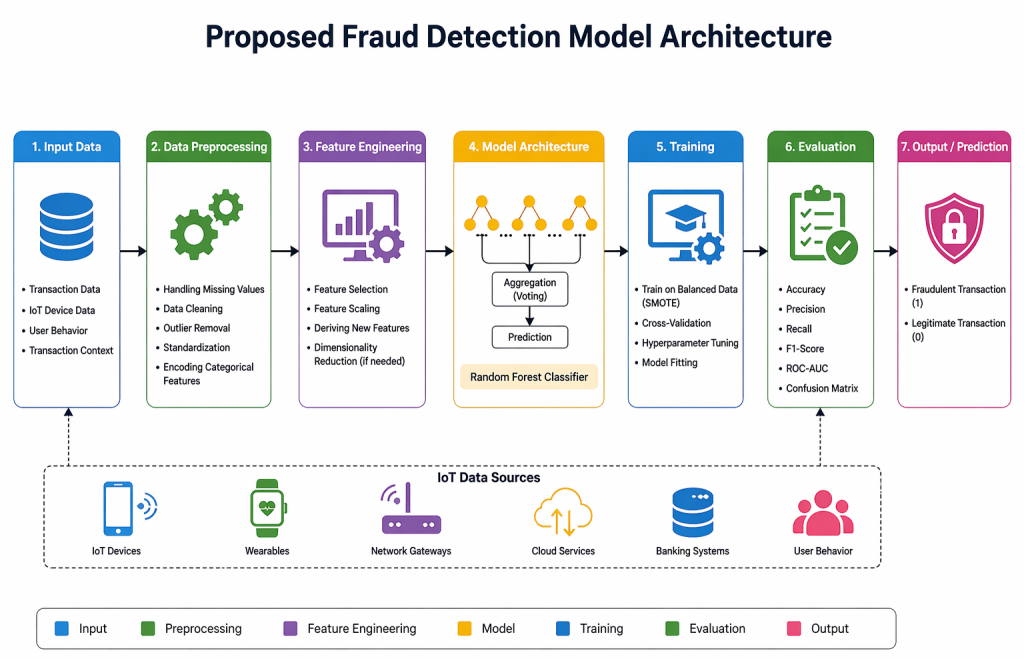

Figure 1: Proposed Fraud Detection Model Architecture

3 Model Implementation

3.1 Selected Model

Random Forest Classifier

Random Forest was selected because it performs well on imbalanced classification tasks, captures non-linear fraud patterns, reduces overfitting, and is widely used in financial fraud detection.

3.2 Hyperparameters

```
RandomForestClassifier(
    n_estimators = 100,
    random_state = 42,
    n_jobs = -1
)
```

3.3 Tools and Libraries

- Python

- Jupyter Notebook
- Pandas
- NumPy
- Scikit-learn
- Imbalanced-learn (SMOTE)
- Matplotlib
- Seaborn

4 Evaluation Metrics

Table 3: Model Performance

Accuracy	0.9994
Precision	0.8265
Recall	0.8265
F1 Score	0.8265
ROC-AUC	0.9131

4.1 Confusion Matrix

```
[[56847    17]
 [   17    81]]
```

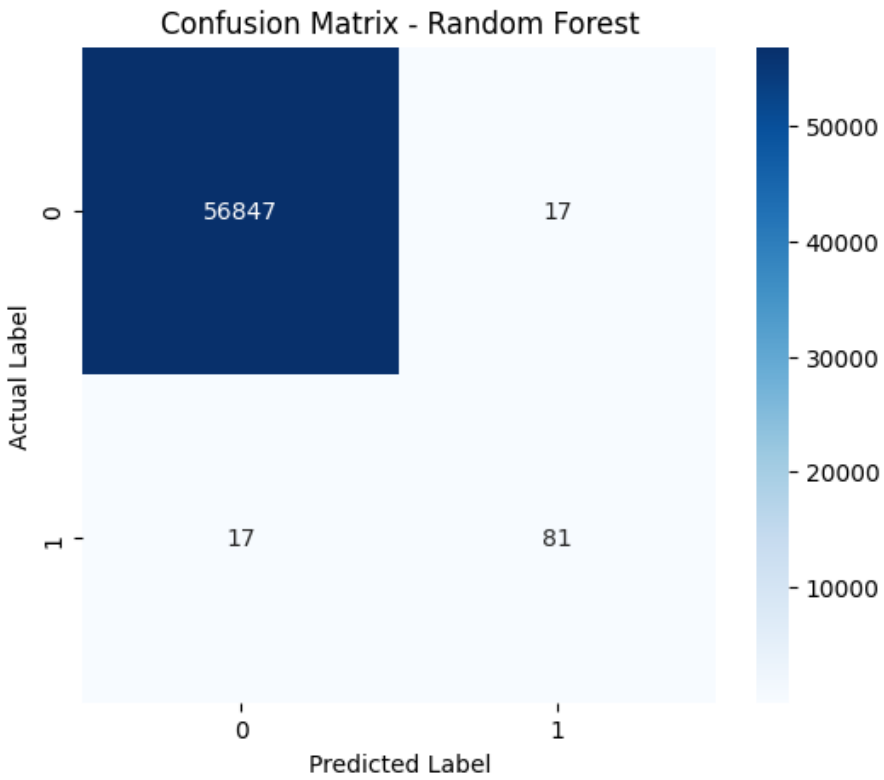

Figure 2: Confusion Matrix Heatmap

5 Results and Discussion

The model achieved excellent fraud detection performance. It correctly identified 81 fraudulent transactions while only 17 fraud cases were missed.

5.1 Strengths

- Very high overall accuracy
- Strong fraud detection recall
- Low false positive rate
- Effective handling of class imbalance using SMOTE
- Reliable performance of Random Forest on banking data

5.2 Limitations

- Fraud cases remain difficult due to rarity

- Some fraud cases are still missed
- Real-time IoT streaming was proposed conceptually, not implemented directly in dataset form

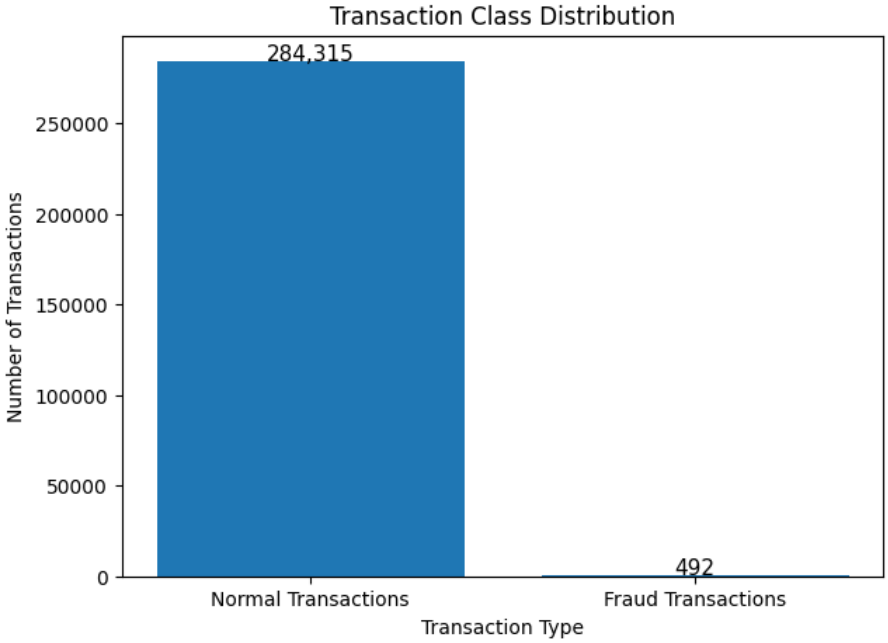

Figure 3: Transaction Class Distribution

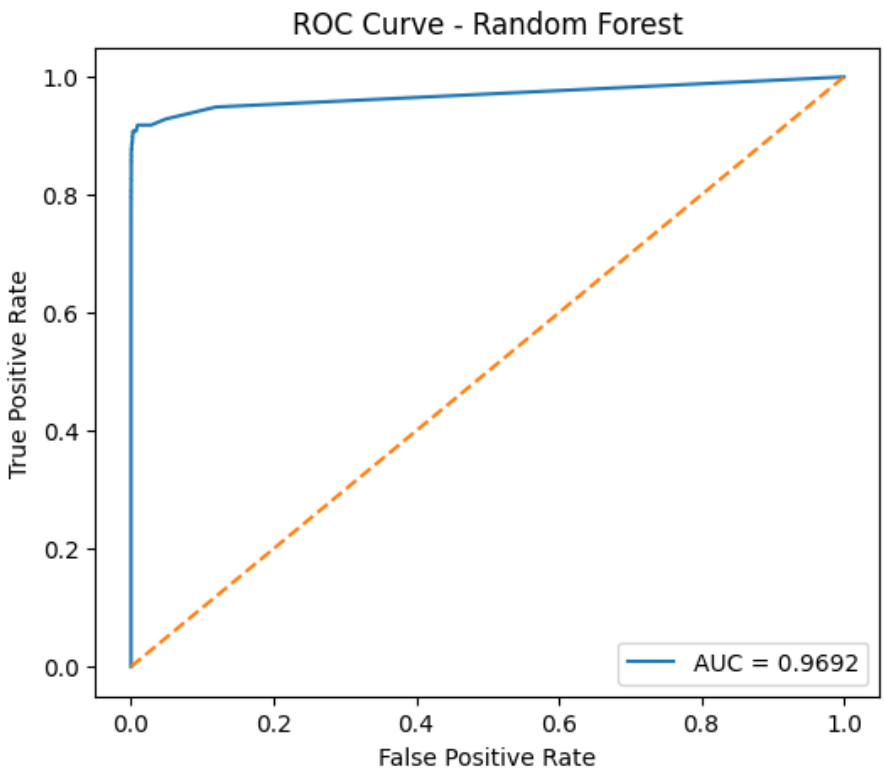

Figure 4: ROC Curve

6 Comparison with Existing Studies

Table 4: Comparison with Previous Studies

Study	Dataset	Model	Accuracy
Taha & Malebary (2020)	Credit Card Fraud Dataset	LightGBM	99.2%
Alarfaj et al. (2022)	Credit Card Transactions	Deep Learning + Ensemble	99.4%
Alatawi (2025)	IoT-Based Dataset	Random Forest	99.99%
Proposed Study	Credit Card Fraud Dataset	Random Forest + SMOTE	99.94%

6.1 Research Gap

Most previous studies focus only on transaction-level fraud detection. Very limited research combines IoT-generated behavioral data, customer risk prediction, MIS reporting, and real-time fraud prevention together.

7 Conclusion

This project successfully implemented a machine learning-based fraud detection system using the Credit Card Fraud Detection Dataset.

The major contributions include proper preprocessing, class balancing using SMOTE, Random Forest implementation, strong evaluation using multiple metrics, confusion matrix analysis, and comparison with existing studies.

The final system shows that Random Forest with SMOTE is highly effective for fraud detection in retail banking and provides a strong foundation for future IoT-driven real-time fraud prevention systems.